ActInf Textbook Group ~ Cohort 2 ~ Meeting 2 (Chapter 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | *Cohort 2 ~ Meeting 2 (Chapter 1)* | *58:47*

So, okay. Hello, everyone. Thanks for joining this second meeting of the second cohort of the Active Inference textbook group, where we're going to be discussing chapter one. Today is September 9th, 2022. So I'll minimize this. And you guys can click on my screen and that will show you the full screen of what I am presenting. So what we'll do in the live meetings here for the next two and a half months or so is totally going to be driven by the material that's in the chapters and in the appendices. So we're going to go to the questions page and we're going to look at the questions that have been asked for that week for that theme. And we're just going to start with the most upvoted questions. So here you can upload a question like this by clicking thumbs up. So we will start with the most upvoted questions. And then we can take notes here and the discourse in this answers and discourse. And if you just click on that, it opens this page and anybody is free to write here. And after addressing the questions, which is probably going to be most of the time, and discussing them, then we will move on to maybe look at the ideas more broadly over here.
So we don't really have too many questions. If anybody wants to jump in and write a question, we did send an email reminder that because this was pretty empty over the course of the week. But hopefully thinking by writing in this way will be a structure where we'll be able to hear everyone's perspective and leave the coda in a better place than we found it. So if anybody wants to jump in and write additional questions in the coda, now would be a good time. And then after looking at the questions, we will pull back and focus a little bit more broadly. So maybe we'll go over here to touch on the ideas and insights of the chapter, and then ideas for that week. And then in the last minutes, we can just see if there's been any changes to the project ideas and start to see what people are excited about applying active inference to. And then we will end at the top of the hour.
So let's just head over to the questions, the cohort two questions page, and take a couple minutes to put some questions in here. So just so you guys know, control plus and control minus are good ways to scroll in and out to get like the right view of the screen. And then you can hide this little textbook bar over here. So I'm going to hide that and just give everybody maybe like five minutes to put in a question or something that would maybe like to discuss from chapter one of the textbook.
So
Thank you.
Thank you.
Thank you.
Thank you.
Thank you.
So maybe we could just take like one more minute and wrap up these questions.
Thank you.
Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

I just want to say like to everyone at any point that there's always the affordance of just listening and also of just rewatching this recording.

There's also the affordance in gather to raise your hand.

Let's see if I can show where that is.

It's been a long time.

Over here.

Over here.

There it is down at the bottom.

So there's this raise hand feature and lower hand.

Yeah.

Let's see.

Yeah.

Okay.

Let's see.

And then if you would, you can raise your hand if you would like to share just on that particular question.

We're going to have them up in the coda.

And you can also pull up the answers in the discourse and that will pull up the focus view of the question.

And everybody is free to collaboratively like edit the answers here in the discourse.

It's better to really add and be noisy, even if it's just like a bullet point list or an incomplete thought, or if there's incoherence or if it's contradictory, those are all fine.

It just adds information and perspective on the question.

So, yeah, we're going to open up each question, the answers in the discourse and read what has been read and already contributed in the answers.

And then also people want to share other thoughts that they're having.

That's awesome.

And then if anybody also just wants to be taking notes, just things that are coming to mind or they don't want to share in a speaking way or just taking some notes on what other people are saying.

So this is really a participatory group.

And so we it's just is what we make it here.

Okay, cool.

I'm still leaving this little window open.

I won't see any raised hands and gather.

All right, let's see.

I need a double screen.

All right, let's see if I can do it in the small way.

Okay, cool.

All right.

So let's get into it.

This first question, I'll pop it open.

And it says, what is the difference between a stimulus response mapping and a state action policy?

So I don't know.

Does anybody have any ideas here?

I'm not.

I think that they're both kind of like an if this, then that.

But they're specific, I think, to machine learning.

So if anybody has any of like more depth or background in this area, that would be awesome to hear.

So I don't know the way I kind of think about it.

Oh, is that Arun?

Did you raise your hand?

Yeah.

Can you hear me?

Yeah, cool.

I can give a very I can give an attempt at an answer to that, which from my understanding, please tell me if I'm wrong.

When you've got the stimulus response mapping, you've got 100% certainty on what the stimulus is.

So with the states, you're then factoring in beliefs about you've got uncertainty.

So maybe you just, you know, in a simple form, you could be chucking a Gaussian distribution on every stimulus and then a Gaussian distribution on what your response is to go from state act to go from stimulus response mapping to state action policy.

That's my intuitive understanding of what's going on.

But again, please tell me if I'm barking the wrong tree.

Yeah, that's awesome.

Thanks for sharing.

And I kind of think about it in the same way.

Like, I don't know.

I mean, this is probably totally wrong, but like I think like about the Roomba, like the little vacuum that cleans your house.

And so like if you turn it on, it turns on and moves like that's a stimulus like you push the button and then it turns on and then like it moves around because you pushed the power button like it's a very clear thing

versus like the state action.
Like if it's it's full, like if the vacuum gets full, then it like dumps out.
But like that state of fullness is like a right like a distribution, right?
Like so it can be like more or less full and it senses how far it is away from the dump out bucket or whatever.
So that's like the state action, like the state can be full.
But like what like what is too full for the Roomba?
Like is it a certain weight that's in there or like a certain volume or sensor?
I don't know.
I have I have uncertainty about when the Roomba dumps itself out.
Cool.
Awesome.
So we can just write this in here.
So and anyone is free to take notes and type in here.
So we can make this in here.

So we can make this in here.
So we can make this in here.
So we can make this in here.
So we can make this in here.
So we can make this in here.
So we can make this in here.
So we can make this in here.
So we can make this in here.
So we can make this in here.
Okay, cool.
All right, so yeah, and anyone's free to just get in here and jump in and do this.
All right, so let's look.
Let's look at this one.
All right, so this question says,
is the brain trying to maximize qualia divergence and minimize encoding length?
Ali?
Sorry, I raised my hand before.
I just wanted to add some comments about the previous question
because I think it's a very important one.
You see, as we go on,
specifically in the Chapter 4,
when we encounter the concept of Markov blanket,
I think we can see kind of more granular differences

between those two concepts,
between stimulus response mapping and state action policy.
But briefly, states are just representations of internal or external world.
Right?
And actions are specifically goal-oriented responses.
And they just try to change the state of the system or the agent.
But when we're talking about just stimulus response mapping,
I mean, response doesn't necessarily tries to change the state of the agent.
As an example,
if you touch a hot object,
the pain you feel is just a response, right?
But if you move away your hand from that hot object,
it's an action which tries to change the stimulus.
And one other important distinction here is
the term policy is used in a different sense in active inference literature
literature as compared to reinforcement learning literature.
Namely, in reinforcement learning,
this term is often used to refer to singular mappings
or one-to-one mappings from states to actions.
But in active inference literature,
policy usually refers to possible sequence of actions,
not just singular actions.
And so that's, I think,
one important distinction we need to have in mind.
Cool. Awesome.
Thank you very much.
And I looked that up, actually.
So here, like,
I think that when they talked about state action policies,
they were using it in the machine learning case
because it says in the textbook,
the goal-directed character of action and active inference
is in keeping with early cybernetic formulations,
but distinct from most current theories that explain behavior.
Oh, sorry.
Someone's talking.
Could you please mute?
Thank you.

So it says distinct from most current theories
that explain behavior in terms of stimulus response mappings
or state action policies.
So I think that this here,
they're not talking about the active inference state action policies,
but they're using it in the reinforcement learning way.
So it says also stimulus response or habitual behavior
then comes a special case of a broader family of policies
in active inference.
So I think that the way that they were talking about it
in the textbook refers to the reinforcement learning way
as opposed to the active inference way,
which Ali is correct.
It's definitely different.
Any other comments or thoughts on this question?
And thank you for whoever's taking notes in the coda.
That's always super helpful.
Okay.
So we can move on and look at this question.
Is the brain trying to maximize qualia divergence
and minimize encoding length?
Who has some thoughts here?
Okay.
The idea of maximizing qualia divergence seems like
at some point that's beyond schizophrenic maybe.
So there's like some maybe threshold at which
you would not want to go beyond.
And that has, seems to have something to me,
seems to have something to do with epistemic value.
Like you can't just maximize,
try to maximize your epistemic value forever.
You would, into a pragmatic problem.
It's encoding length.
So I'm just, I guess I'm wondering if maybe a,
a different framing of the question
or commensurate framing of the question is
what's the relationship between epistemic value
and pragmatic value?

Like are there trade-offs there?
Yeah.

I scanned through chapter one and really didn't see
a qualia divergence or encoding length
there in the chapter.
So I'm not really, yeah.
Is this anyone's question?
Do you want to unpack this question
a little bit more for us?
Maybe?
No volunteers?
Awesome.
Daniel, thanks for joining us.
I will, I can officially pass the baton off to you
and maybe you have some comments
about maximizing qualia divergence
or minimizing encoding length here.
But we got in and we wrote a few questions
and we went through the first one.
We could also go back and touch on that
if you've got comments on that as well.
That question just makes me wonder
what is being minimized?
What is being maximized?
And when is it some particular value
that's being minimized or maximized?
And when is it the divergence
between which other differences?
And in chapter one, there aren't,
any of the formalisms yet,
but we'll see a lot of double lines
and KL divergences that will cue us
towards where a divergence is being
at least described.
Cool.
If you're ready, I can unshare my screen
and you can take over.

Please continue.
Okay.
Any more additional comments here?
I like this.
Is qualia, I didn't get to write this down,
but Brock had mentioned.
Is qualia divergence, schizophrenia,
beyond schizophrenia, that was the term.
diversity you can imagine is necessary
and some more might be really healthy and useful
at some point.
I don't, then you're,
you have multiple personalities or something.
I don't.
that makes me think of the sense of normalcy
and what is experienced as normal
and how much of a,
how much of a deviation from normal is perceived as,
like,
what, what,
what is perceived as normal
and where is it perceived as a creative
and maybe even enjoyable novelty,
like a guitar solo?
Like a guitar solo?
And then where is the qualia divergence
too high and maybe uncomfortable?
And then what's the view from the inside
with that actual experience?
Like what qualia brings in?
And then where,
where is this being modeled from the outside
with behavioral descriptions
where we,
we might not be able to directly address
what the qualia are.
And Daniel,
that reminds me like some of it might be,

you said some of it might be normal
and some of it might be uncomfortable
and,
or some of it might be,
some divergence might be creative
and some might be uncomfortable.
And there's actually like a huge degree
of,
of like mental disorder
associated with creativity.
Like,
I mean,
studies have been done
and it's like,
I mean,
you're like,
I mean,
there's the classic,
like,
you know,
songwriter,
singer who commits suicide
and we hear this again and again,
but there's actually like a great degree
of mapping between the creative type
and some kind of atypical mental state.
So maybe both of those are the same.
Okay,
cool.
Anybody have any comments?
I don't know if I can see everybody
in my tiny screen.
Go ahead.
Sorry.
Hello.
Hello.
Hi.
It's not clear to me

what really,
what precisely means
a quality divergence
because divergence
with respect to what?
I mean,
usually a KL divergence
is a divergence of,
like a measure
of the difference
between two distributions.
So here,
what does exactly mean?
Maybe the person asking the question
and clarify.
I'll also just note that,
oh, go ahead, Brock.
Sorry.
I was just,
I don't,
I don't know if they're here.
I think it's,
who already asked,
but to me,
it just seems like
an internal kind of
population of one qualia set,
another qualia set
that are differentiable.
Like,
you can have divergence
certainly of qualia
in your own experience
if you just consider,
you know,
the various sensory
kind of experiential states
that you,

you occupy.
Like,
okay,
so that means,
that would mean,
like,
expanding the range
of,
of your
phenomenology,
something like that.
Yeah,
it could be.
Yeah.
So,
yeah,
so you could imagine
a sense of space
and self,
you know,
that's kind of
frequent
and normative
that if that starts
to,
maybe in a religious
or spiritual setting
or other kind of
group team settings,
you know,
you kind of get a group flow
going maybe,
like that,
that starts to change.
But,
but maybe
if those two things
greatly diverge,

that wouldn't be great.
So,
that maybe is related
to the
complexity
of that part
of the model
that encodes
a representation
of the self.
So,
that would increase
the complexity
of the,
of the model,
which is one part,
one part
of,
of the free energy
minimization scheme.

the,
I guess,
I mean,
with each increase
in the epistemic
import
of,
of the,
of observation,
then,
also,
complexity
of the model,
though,
needs to be kept
at,
within,

within some limit.
So,
inside your blanket.
So,
yeah,
so it's,
probably,
it's,
it's a trade-off
between,
increasing the complexity
of the model
for what concerns
the range
of possible
experiential,
the,
yeah,
qualia,
and their epistemic value,
I guess.
I'll also just mention,
out of fairness,
that qualia is not mentioned
in the textbook.
Mm-hmm.
But it's totally awesome
to consider
what the formalisms
entail
for philosophy
and many papers
and many discussions do.
And it's like,
an important area
to explore.
But,
it's also not

what is,
addressed.
And I,
I think it's probably,
even though these authors
in other settings
tackle the problem
head on,
the textbook
is
seemingly crafted
to enable
some discussions
that might not
require
the consideration
of the view
from the inside
or the phenomenology
that qualia
entail.
And that
helps the generality
of the,
and the applications
across different systems.
Mm-hmm.
And so,
understanding how they map
onto our
undeniable experience
is a great
area to explore.
Yeah,
overall the-
Sorry, go ahead.
No, no,
I was saying that

overall the book
doesn't,
doesn't expand
too much
on the,
on the part
of the model
that represents
the,
the agent
itself
as a,
you know,
as an entity,
like,
you know,
our own self-representation.
It,
it mentions it
but does not
really,
does not really
dwell onto it.
More about that
is probably
in the book
of Jacob Howe.
There's a little bit
more on that
but
in this one
not so much.
Well,
perhaps this connects
to the
following question
on self-evidencing.
So,

before we jump
forward,
I like,
I just want to
stay on this
question for one
second.
So,
I looked up
the definition
of qualia here
and it's the,
the individual
instances of
subjective conscious
experience.
And I don't know,
maybe,
in the question,
which let's read
it again,
it says,
is the brain
trying to maximize
qualia divergence
and minimize
encoding length?
So,
I was wondering
if this maybe
is meant
for a question
like in an
evolutionary sense?
Like,
is this what
the brain
is trying to do

across time?
Like,
in deep time?
From,
you know,
prehistory to now?
So,
yeah,
we can move on
though
because maybe
self-evidencing
is,
is more
addressed in the book
than
qualia divergence.
questions.
Yeah.
So,
what do
self-evidence
and inactivism
really mean?
Does anyone
want to
take a stab
here?
Yeah,
I asked
the question
because when
I read
up,
my eyes
just glaze
over when
I hit these

words.
I can read
them in sentences
and I just,
just doesn't
stick in my
head.
What is really
meant,
self-evidencing,
just,
it just
turns into
something
tautological.
I don't
have any
sort of
real grasp
of what is,
what is being
said in these
sentences.
I can take
a stab
at that
because
really
it's a,
it's a,
it's quite a
simple concept.
So,
if we think
about
the agent
not as
having

a model
of the
world,
but being
itself
a model
of its
own
environmental
niche,
then
minimizing
free energy
basically
means
working,
acting,
so as
to
collect
the
observation
that it
expects,
that is
minimizing
surprise.
So,
since
you can
say
finding
the
evidence
for its
model
of the
world,

but since
the agent
is a
model of
the world,
that becomes
self-evidencing.
So,
basically,
you could also
see it
as trying
to find
the,
the sensory
stimulation
that is,
that makes
your phenotype,
your organism
viable,
that sustains
you.
So,
that's,
that's,
evidencing
meaning finding
the observation,
the sensory
evidence,
that is more
coherent
with your
makeup as an
organism.
thanks.
I'll,

I'll add a
few notes.
So,
inactivism is
a long-lived
thread in
cognitive
sciences that
is long
existing before
active inference,
and it is
describing
that systems
with bodies,
real,
particular
things,
existing in
the world
ought to
be having
their cognition
modeled as
bodies engaging
with the
world,
rather than
like them
taking like
a view from
the outside
or necessarily
needing everything
to be like
symbolic
communication.
Whereas in

digital systems
or kind of
abstractions of
cognitive systems,
it's very easy
to get into
some information
and symbol
passing
discussions.
Whereas for
real systems
with biomechanics
and embodiment,
inactivism brought
attention to
the requirement
of considering
like the
active engagement
of an entity
with its niche.
So,
maybe it's not
a term that
provides a
huge cognitive
model update
because for many
people it's
very natural
to consider
cognitive systems
as inactive
and embodied,
embedded,
and cultured,
et cetera,

et cetera.
And sometimes
that's called
4E cognition
and those are
all in allegiance
together.
But that's what
inactive points
towards.
And self-evidencing
is describing
that the
imperative
of systems
as modeled
under active
inference
is for them
to accrue
consistent
observations
and find
evidence
for the kind
of thing
that they
expect and
prefer themselves
to be.
And in the
coming weeks
we'll explore
this like
expect and
prefer and
how can
expectations

be preferences
and what about
when we expect
bad things to
happen and
uncertainty
about expectations.
But we can
contrast this
imperative for
finding
consistent,
self-consistent
or self-evidencing
observations
rather than
pursuing
rewarding
observations.
so one
can say
I expect
and prefer
to be
at room
temperature
and then
moving around
the space
or putting
the jacket
on or off
to reduce
surprise
about what
observations
one is
receiving

as opposed
to needing
to construct
a reward
framework
in which
room temperature
is rewarding
and then it
gets less
rewarding
and then
what is
being
maximized
is the
reward
of the
cognitive
system
and again
in contrast
self-evidencing
puts the
imperative
on the
system's
self-model
and reducing
surprise
about its
own observations
rather than
seeing those
observations
as just
proxy
of reward

states
that have
to be
maximized.
Arun?
I found
both of those
answers really
interesting
and I have
a question
to each
of those.
So on
the self-evidencing
we have
so maybe
this is fast
forwarding a
little bit in
the book
but they talk
about both
generative models
and generative
processes
and those
are distinct
ideas.
So when we're
talking about
an organism
as a model
in its niche
it's still
distinct from
the generative
process, right?

So that's
something that
I'm not
100% sure
on and I
also find
the self-evidencing
part a little
bit circular
and I don't
quite have a
good understanding
of that.
So in my
mind the
generative model
and the
generative process
are very
distinct and
yes you're
looking for
you're an
organism and
you're in
your niche
and you want
to succeed
and by
natural selection
or whatever
we only really
see organisms
that succeed.
They're going
to look for
things that
meet their

priors such
as you know
a nice
homeostasis
food
reproduction
all of that
sort of stuff
that's sort
of my
understanding
of that
part
but it's
still an
organism
using a
model to
navigate the
world
rather than
it is a
model itself
and then
also on the
idea of
minimizing free
energy but
also it's
not a
reward.
To me it
sort of is
a reward
it's just a
reward with
respect to a
prior distribution

so yes you're
minimizing free
energy and
you have a
prior preference
of being at
room temperature
that's great
but you'll
still have an
objective
function or
functional I
think as the
book calls it
you're just
subtracting the
prior distribution
of temperatures
that your body
likes.
So I still see
that as like
it's fundamentally
an optimization
problem it's just
the units are a
bit different and
it's all statistical
distributions and
the differences
between them
rather than I
don't know x
minus x bar
squared that sort
of thing.
So those are my

those are the areas
that I'm not very
clear on so I
don't know if
anybody can clear
those up.
Thank you.
So can I
follow up on
that then?
So on the
inactivism I
think the
problem is
knowing what
the specific
meaning of it
is compared
with embodied
and embedded
because those
those E's are
so often used
together.
What is
where would
you say
inactivism
rather than
embodied?
and I
think you
touched on
it by saying
about it
involving having
having a model
maybe whereas

being embodied
doesn't.
I would
suggest that
inactivism
calls the
action into
focus whereas
the embodiment
and the embedded
refer to just
the materiality
but the reason
why they are
allegianced as
for EEEE
etc.
they're just
fingers pointing
at the moon
they're just
fingers pointing
at real
embodied
organisms
and some
of the
complexities of
modeling them
and framing
them in
contrast to
disembodied
approaches.
So
they're pointing
at organisms
and they're

different theoretical
vectors that have
been taken
and some of
the synthesis
amongst them
has already
been carried
out.
Yeah.
An activism
was
became
popularized
the notion
with Francisco
Varela
by Francisco
Varela
actually
and
I think
he was not
too fond
of the term
actually
but
anyway
he
the thing
is that
he wanted
to stress
the fact
that cognition
and consciousness
emerges
in the

circularity
between
sensation
and action
so
that was
enacted
in the
circularity
continuous
circularity
of
action
and sensation
so
I also feel
it's
really close
to
the free energy
approach
and active
impressions
although
there are
some of
the early
champions
of an
activism
like
Evan Thompson
and
some other
scholars
that
argue
that

there are
actually
crucial
differences
anyway
for what
concerns
the
the
difference
between
the generative
process
and the
generative
model
the generative
process
is basically
the actual
physical
processes
that turns
causes
into
stimuli
into
that
impinge
on our
sensory
surfaces
the
generative
model
is the
best
guess

that
an
organism
can
make
about
that
so
they're
not
identical
of course
the
organism
tries
to learn
at its
best
the
generative
process
but
you know
it's just
a model
Ali
yes
and
I also
wanted to
mention
Jakob
Howey's
well-known
paper
the
self-evidencing
brain

which I
believe
is the
seminal
paper
which
introduced
this concept
of self-evidencing
to active
inference
literature
and
in that
paper
he also
contrasts
this term
with
inactivism
and activist
and all
the other
four
e-cognition
frameworks
and theories
but
briefly
he believes
that
the
self-evidencing
brain
by putting
a boundary
namely
the Markov

blanket
as
one of its
essential
components
in order
for it
to be
able
to infer
the states
of the
environment
it can
retain
its
cognition
within
its
inner
statistical
or generative
model
so
that's
one of
the reason
the
self-evidencing
concept
was deemed
necessary
to introduce
at the
first place
so
and
self-evidencing

is something
that
just
is
beyond
the
biological
brain
and
it
can
refer
more
generally
and
more
broadly
to
any
agent
which
tries
to
learn
about
its
environment
as a
one
dynamical
system
which tries
to perform
inference
about
the
other
dynamical

system
which is
the
outside
world
but
with
a
particular
aim
of
trying
towards
minimizing
the error
or
performing
the
allostasis
so to speak
so
that's
one of
the
main
differences
between
inactivism
and
self-evidencing
concept
here
it's also
very
thank you
all for sharing
it's very
illustrative

to look at
figure
1.2
and see
where these
terms
are used
in the
context
of the
low road
and the
high road
to active
inference
which is
one of
the
key
concepts
in
this
chapter
and
it's
describing
how
the
following
chapters
are
going
to be
structured
so
we
see
self-evidencing

on
this
upper
branch
of
the
subway
system
coming
down
from
the
high
road
and
we
see
some
of
these
other
terms
that
people
have
been
mentioning
like
generative
model
coming
up
from
the
low
road
and
then

in
the
same
page
that
we
were
just
discussing
I
copied
some
sections
from
the
text
that
are
describing
these
high
and
low
road
they
describe
that
the
high
road
is
starting
from
the
question
of
how
living

systems
persist
and
act
adaptively
in
the
world
and
so
that
is
very
resonant
with
self
organization
which
is
the
first
stop
on
that
subway
system
from
the
high
road
and
also
Varela's
work
autopoiesis
a lot
of
questions

about
how
cognizant
sentient
systems
persist
the
high
road
perspective
is
useful
to
understand
what
living
organisms
must
do
and
why
why
to
be
persistent
via
surprise
minimization
not
via
reward
maximization
and
then
what
they
have
to

do
so
it's
the
what
and
the
why
in
response
to
the
broader
question
of
how
complex
systems
persist
that's
the
high
road
and
we've
had
many
discussions
and fun
for
every
person
to
also
think
about
this
high

road
low
road
distinction
but
the
high
road
is
starting
from
a
complex
system
thriving
or persisting
in a
complex
niche
we can
think
of
and
then
the
low
road
is
starting
from
Bayes
theorem
and
continues
up
through
the
Bayesian

brain
also
motivating
active
inference
from
a
different
direction
and
it
is
describing
the
low
road
perspective
is
useful
to
illustrate
how
active
inference
agents
minimize
their
free
energy
well
they're
not
necessarily
doing
frequentist
statistics
they're
doing

something
that we
can
describe
as
Bayesian
and
this
textbook
is
going
to
introduce
Bayes
theorem
briefly
in a
box
and
it's
a
big
area
and
some
people
will
be
familiar
with
thinking
about
priors
and
updating
priors
and
the

way
that's
discussed
in
Bayesian
statistics
some
will be
familiar
for some
it might
be
novel
but
this
is
describing
kind
of
the
atomic
computational
or
statistical
nucleus
that is
going
to
describe
how
different
systems
can
fulfill
that
imperative
that we
saw

traced
out
by
the
high
road
and
Ali
I know
you'll
have
a lot
to
add
on
that
I've
also
put
a
PDF
to
Carl
First
essay
Beyond
the
Desert
Landscape
on
resources
page
and also
on
the
chapter
one
questions

and notes
page
I
believe
this
essay
is
the
most
comprehensively
articulated
and
in my
opinion
very
beautifully
even
poetically
articulated
account
of the
high road
and low
road
distinctions
and the
philosophical
distinctions
between
them
if you
scroll
down
I think
you can
find
the
paper

I've
attached
there
yeah
right at
the bottom
of the
page
and
I've
also
appended
yeah
that's
it
the
Carl
Christen's
no
beyond
below
that
yeah
beyond
the
desert
landscape
and
I've
also
appended
Andy
Clark's
reply
to this
essay
because
this

chapter
comes
from
this
book
called
Andy
Clark
and
its
critics
which
I
think
can
help
to
understand
the
real
differences
between
those
two
concepts
and
why
it's
important
to
distinguish
between
them
when
we
want
to
discuss

and
understand
any
active
inference
related
concepts
thank you
awesome
thanks
I'm
going to
just
make
note
of
that
in
here
maybe
just
the
comment
on
the
why
if
you
Daniel
like
talking
about
the
high
road
that
is
it

why
really
because
my
understanding
really
is
the
perspective
or the
nature
of the
free
energy
principle
is
really
as
Carl
says
is
more
tautological
than
teleological
and
why
since
the
point
to
some
sort
of
teleology
so
it's
go ahead

go ahead
yes
scientific
or formal
addressments
of teleology
or end
or goal
directedness
are very
interesting
and
another
framework
or
theory
which is
evolution
has also
threaded
the needle
with
tautology
and
teleology
and
so
there's
a lot
to be
said
there
about
how
those
two
categories
aren't

necessarily
disjoint
do you
believe
they have
to be
disjoint
or are
you
expecting
or preferring
that they
are
if the
imperative
is to
persist
and survive
and we
simply
don't
observe
systems
that
don't
then
that
phenomena
can be
both
tautological
in the
sense
that
it's
self
describing
as well

as
teleological
in that
from a
perspective's
view
not speaking
for the
system
itself
but from
our view
on the
system
it is
acting
as if
it is
trying
to
survive
the bubble
is acting
as if
it's
trying
to get
to the
surface
of the
water
in the
sense
that it's
not
acting
like it's
trying

to go
deeper
we don't
need to
entail
any
qualia
or any
sort of
broader
theological
reason for
that to
occur
we can
describe
something that
is both
teleological
and
tautological
yes
no I
think I
agree
I think
that
from what
from what
Carl
often says
that it's
you know
deflationary
accounts
is that
basically
the

the free
energy
principle
basically
basically
describes
self
organizing
system
just because
it persists
and so
what
characteristics
does it
need
in order
to persist
and
so
in that
sense
is
tautological
so
it's
just
you know
if it
persists
it must
behave
like
that
so
you don't
even need
to

put in
any
imperative
to survive
if it
survives
it must
behave
like that
but of
course
as you
said
you
can
frame
this
self
organizing
drive
in a
sense
as
some
sort
of
imperative
to
maintain
its
form
and
function
of
the
organism
so
yeah

there's a lot
of interesting
things to say
about that
and
also
one
note
is
sometimes
we speak
in a mode
of like
what the
imperative
is for
the system
as
itself
as
what it
normatively
should
or
imperatively
must
be doing
but also
we can
just
speak
from
ourselves
as
modelers
of
other
things

as
what
our
modeling
imperatives
are
and
that is
a
slightly
different
question
and
that's
sometimes
known as
instrumentalism
which can
be
compatible
with
realism
of
various
flavors
and
it's
been
explored
broadly
however
from
our
perspective
as
observers
things
that are

not
acting
as
to
minimize
their
surprise
relative
to
some
area
in
space
that
makes
us
say
it's
that
kind
of
a
thing
if
it's
not
staying
in
that
space
which
we're
defining
as
that
kind
of
thing

it's
not
that
thing
if
the
tornado
were
not
staying
in
some
range
of
phase
space
that
we
call
tornado
it
wouldn't
be
a
tornado
to
us
and
so
sometimes
that
can
be
a
more
empirically
grounded
and

less
speculative
stance
to have
two
feet
as
us
as
modeling
systems
which
is
going
to
be
consistent
with
the
meta
Bayesian
approach
that's
discussed
later
in
the
book
and
it
sidesteps
or it
preempts
a lot
of
very
thorny
questions

around

speaking

for what

systems

like

actually

are

or

actually

are

doing

and

it's

the

difference

between

saying

the

thing

is a

model

of its

niche

you know

big if

true

we're going

to model

it as if

it's

a model

of its

niche

it's

irrefutable

that's just a

choice that

we're taking

and someone
could say
well I
would have
taken a
different
choice
in
modeling
and you
can say
great
what would
your
choice
have
been
and
there's
actually
a
starting
point
instead
of just
this
endless
speculation
around
what is
the
system
really
and
especially
when we
think
about

our
Markov
blankets
and
our
inability
to make
direct
contact
to
hidden
states
it
bolsters
the
justification
of speaking
as a
modeler
very cool
anyone else
with comments
on this
question
this
self-evidencing
and an
activism
which we
could talk
I'm sure
all day
about
so if I
just
finish off
that
I think

possibly
it's the
language
that's
the problem
with
self-evidencing
it just
sounds as
though
it's
self-confirmatory
all it's
doing is
looking for
support
for its
existence
which
is not a
good strategy
for survival
why you
say that
if I
only look
for evidence
of my
existence
rather than
things that
are a
challenge
to that
then
it
is
possibly

related to
the
dark
room
problem
yes
exactly
yeah
so
so
in the
for the
sake of
long-term
persistence
it may be
necessary
to
incur
sometimes
transiently
into states
that are
surprising
so
there
is no
real
there's no
real
incoherence
there
I think
no
I agree
I'm just
challenging
it's

as a
language
a bit
of a
language
barrier
yeah
great
point
and
indeed
the natural
language
is often
heavily
shading
interpretations
and
people's
conceptions
of different
terms
like
around
surprise
belief
expectation
preference
all these
everyday
terms
which are
core
in the
active
inference
ontology
it's

really
important
to
interrogate
our
understanding
of them
and check
their
alignment
with
formalisms
and
this
book
is going
to a
large
extent
focus
on
the
kernel
of the
active
inference
action
perception
loop
single
agent
single
level
kernel
and
in
that
way

it's
very
analogous
to
a
single
linear
regression
in
practice
linear
regression
on
large
data
sets
does
sweeps
across
families
of
models
and
there's
all
kinds
of
things
that
come
into
play
just
for
the
linear
regression
that's

the
kernel
and
so
in
a
nested
model
one
could
reduce
surprise
and find
confirmatory
evidence
I'm the
kind of
thing
that
seeks
out
novel
information
sources
and
therefore
seeking
out
novel
information
sources
is
confirmatory
evidence
of that
so
once
we

step
into
multi
layer
models
many
of
the
absolutely
valid
limitations
of
the
kernel
in
and
of
itself
are
addressed
so
those
are
like
all
the
directions
that
people's
questions
on
and
written
down
are
super
important
they're

all
the
things
that
are
direct
adjacencies
and relevant
for
applications
and there's
going to
be things
about
real
complex
systems
that
the
kernel
does
not
describe
totally
so
this
isn't
like
the
end
of
the
active
inference
modeling
of
complex
systems

total
book
it
really
is
just
describing
like
the
essence
of
the
loop
so
I
want
to
just
give
Arun
the
last
comment
because
it's
the
top
of
the
hour
now
so
Arun
yeah
I
just
wanted
to

add
to
Neil's
question
again
like
I
fully
understand
where
you're
coming
from
Neil
when
I
first
read
about
the
self
evidencing
thing
I
was
like
that
makes
zero
sense
and
exactly
what
you
described
like
oh
okay

you're
not
going
to
look
for
predators
like
as
a
basic
thing
that's
not
evidence
of my
existence
that's
evidence
for my
death
but I
think
the way
I've
sort
of
come
around
to
it
is
it's
all
about
the
prior
preferences

and
what
you
expect
to
see
and
yeah
this
occupation
of face
space
I think
is quite
important
of
you
sort
of
get
attracted
to
particular
states
where
you're
comfortable
but you
do need
to know
about
the rest
of the
world
anyway
and you
have to
plan

for
not
being
in those
states
sometimes
and so
looking for
evidence
of your
existence
is all
about
matching
those
priors
and those
priors
can be
pretty
flexible
so
yeah
that's
I think
that will
in my
mind
like that
comes
through
a bit
later
on
in the
book
I think
so

I would
say
my advice
to you
is stick
with it
but
it is
it was
very
counterintuitive
to me
as well
when I
first came
across it
yeah